

# Design and Deployment of a Wazuh-Based SIEM Home Laboratory for Threat Detection and Alert Triage

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## Abstract

This paper presents the design, deployment, and evaluation of a Security Operations Center (SOC) home laboratory built on the Wazuh open-source SIEM platform. The laboratory comprises three virtual machines: an Ubuntu-based SIEM server hosting the Wazuh Manager, Indexer, and Dashboard; a Windows 10 endpoint running the Wazuh agent; and a Kali Linux machine used for offensive testing. Attack scenarios including aggressive port scanning, SMB enumeration, network reconnaissance, and authentication failures were simulated to validate detection capabilities. Results demonstrate successful agent enrollment, log collection, alert generation, and MITRE ATT&CK mapping, enabling practice of core SOC Analyst (L1) workflows such as alert triage and threat hunting. Visibility limitations are identified, and enhancements including Sysmon telemetry and custom detection rules are proposed.

Keywords— SIEM, Wazuh, Security Operations Center (SOC), threat detection, alert triage, MITRE ATT&CK, blue team.

## 1. Introduction

The increasing sophistication of cyber threats has made the Security Operations Center (SOC) a central component of organizational defense. Within a SOC, Security Information and Event Management (SIEM) platforms aggregate, correlate, and analyze telemetry from heterogeneous sources to enable timely threat detection and incident response. Consequently, hands-on competence in SIEM deployment, alert triage, and threat hunting is a prerequisite for entry-level SOC Analyst (L1) roles, yet access to production SOC environments is rarely available to students. To bridge this gap, this project develops an isolated SOC home laboratory using the open-source Wazuh SIEM platform [1]. The laboratory replicates a realistic three-tier

environment — SIEM server, monitored Windows endpoint, and attacker machine — used to simulate offensive activity and practice defensive workflows. This paper documents the architecture, implementation, attack simulations, and observations of the laboratory as a portfolio reference for blue-team skill development.

## 2. LAB OBJECTIVES

The principal objectives of this work are:

- (1) deploy a fully functional Wazuh SIEM;
- (2) configure and monitor Windows 10 endpoint;
- (3) simulate real-world attacks from a Kali Linux machine;
- (4) practice alert triage and threat hunting;
- (5) understand log collection, correlation, and visualization in a SOC environment; and
- (6) document the process for professional use.

## 3. LAB ARCHITECTURE

The laboratory was virtualized using Oracle VirtualBox on an isolated host-only network (192.168.56.0/24). The components are summarized in Table I. Telemetry flows from the attack source to the visualization layer as follows:

Kali Linux (attacker) → Windows 10 (endpoint) → Wazuh agent → Wazuh Manager → Indexer → Dashboard.

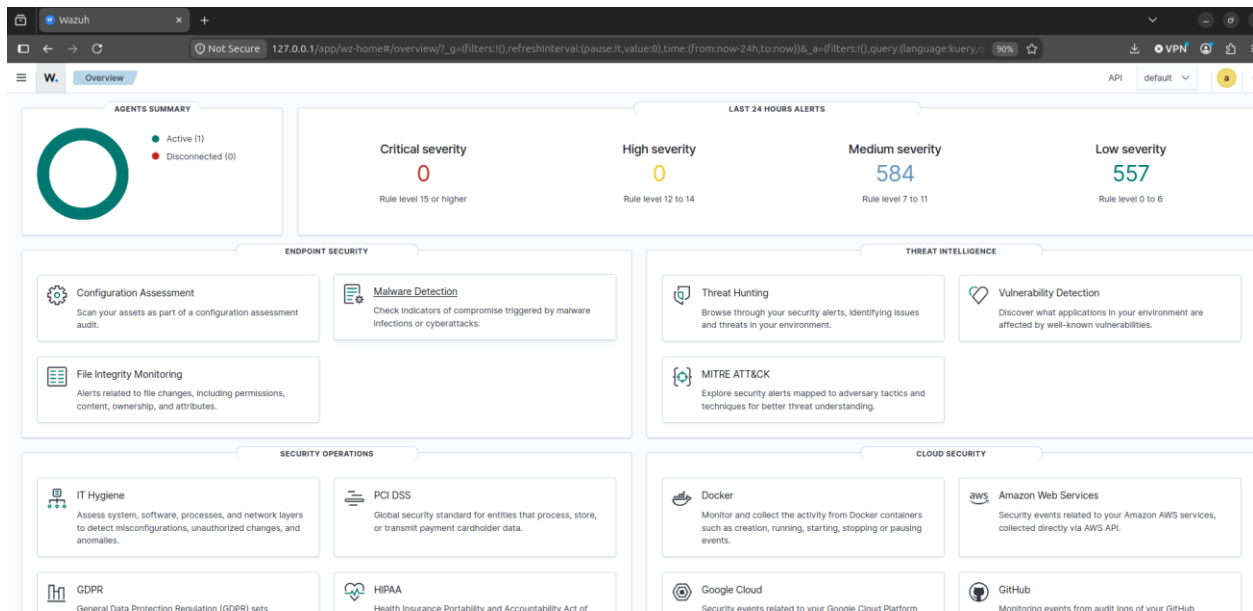
| Component          | Operating System | IP Address     | Role                                  |
|--------------------|------------------|----------------|---------------------------------------|
| SIEM server        | Ubuntu server    | 192.168.56.107 | Wazuh Manager<br>+Indexer + Dashboard |
| Monitored Endpoint | Windows 10       | 192.168.56.106 | Wazuh Agent (log source)              |
| Attacker Machine   | Kali Linux       | 192.165.56.103 | Offensive testing & attack simulation |

TABLE 1 - LABORATORY ARCHITECTURE

## 4. IMPLEMENTATION

### A. SIEM Server Deployment

An Ubuntu Server virtual machine was provisioned as the SIEM server. Wazuh was deployed using the official all-in-one installation script (wazuh-install.sh -a), which provisions the Manager, Indexer, and Dashboard on a single node [2]. Host firewall rules were configured to permit agent communication on TCP ports 1514, 1515, 55000, and 443, and deployment was verified by accessing the Wazuh Dashboard, shown in Fig. 1.

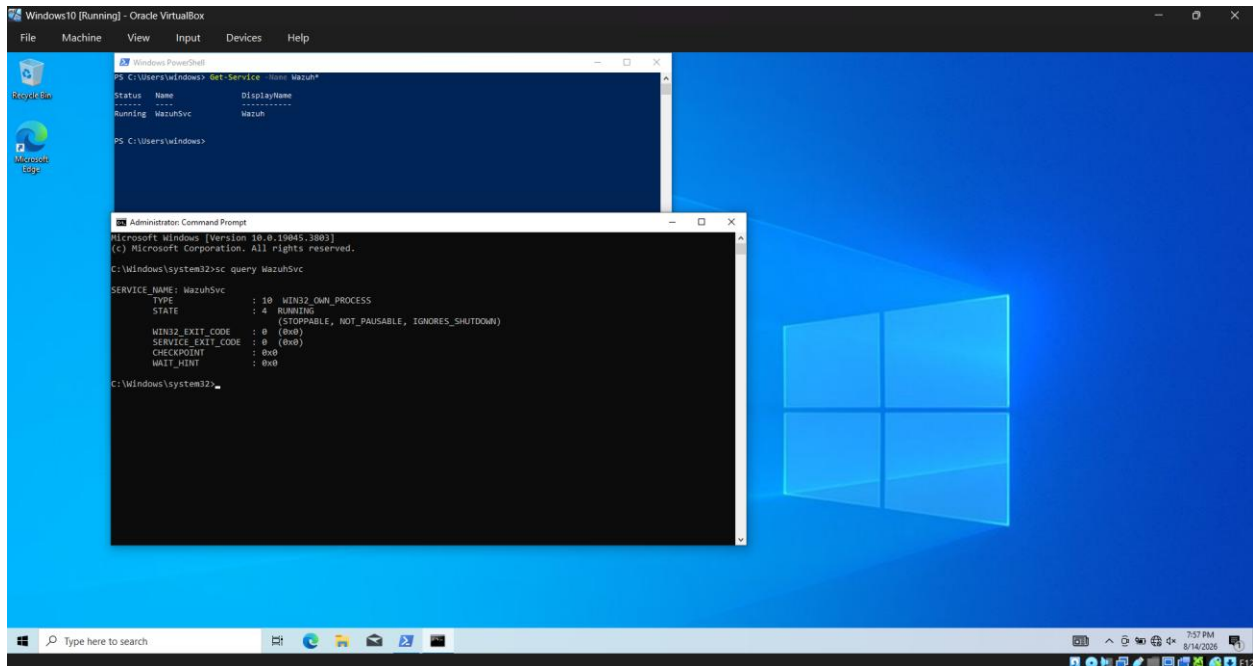


【FIGURE 01: Main SIEM.png】

Fig. 1. Main Wazuh SIEM dashboard following successful all-in-one deployment on the Ubuntu server.

## B. Windows Endpoint Configuration

The Wazuh Windows agent was installed on the Windows 10 endpoint and pointed to the Manager at 192.168.56.107. Successful enrollment was confirmed by the Active agent status displayed in the dashboard, as shown in Fig. 2.



**【FIGURE 02: Windows\_running\_SIEM.png】**

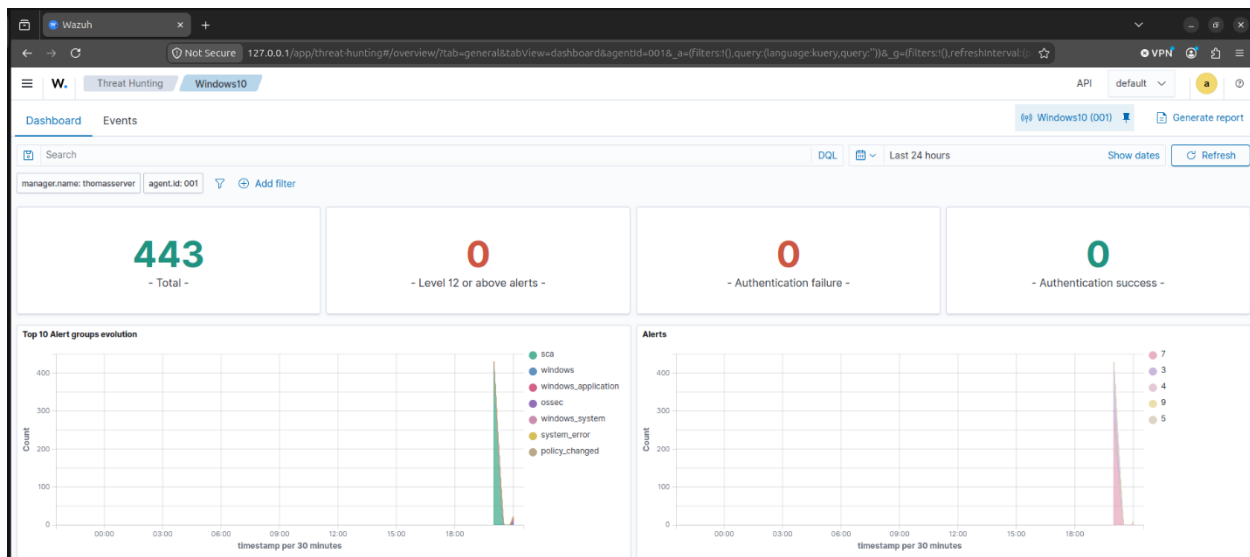
Fig. 2. Windows 10 endpoint with the Wazuh agent installed and reporting to the manager.

### C. Attacker Machine Setup

A Kali Linux virtual machine was configured with a static address within the host-only subnet, with connectivity verified to both the SIEM server and the Windows endpoint. Offensive tooling, including Nmap and enumeration scripts, was prepared for attack simulation.

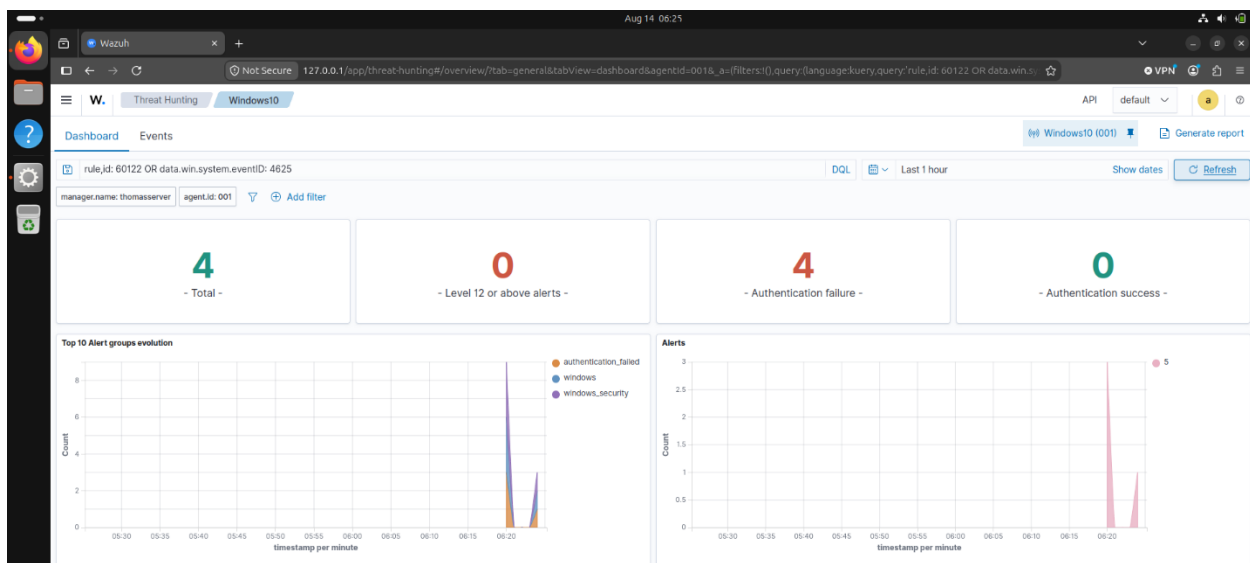
## 5. ATTACK SIMULATION

Prior to launching attacks, a baseline of the environment was captured: Fig. 3 shows the threat hunting module before any malicious activity, and Fig. 4 shows the attacker terminal prior to executing Nmap scans. The attack scenarios listed in Table II were then executed from the Kali machine against the Windows 10 endpoint using Nmap and credential-based tests [3].



【FIGURE 03: Threat\_hunting\_dashboard\_before.png】

Fig. 3. Threat hunting dashboard (baseline) prior to attack simulation, showing no malicious events.



【FIGURE 04: Before\_Nmaps.png】

Fig. 4. Attacker (Kali Linux) terminal before launching Nmap scans against the Windows endpoint.

| Attack Type            | Tools / Command                    | Purpose                |
|------------------------|------------------------------------|------------------------|
| Aggressive Port Scan   | <code>nmap -A -T4 -p 1-1000</code> | Service & OS detection |
| Authentication failure | Invalid credentials                | Detect SIEM logs       |
| SMB Enumeration        | <code>nmap SMB script</code>       | Share & user discovery |

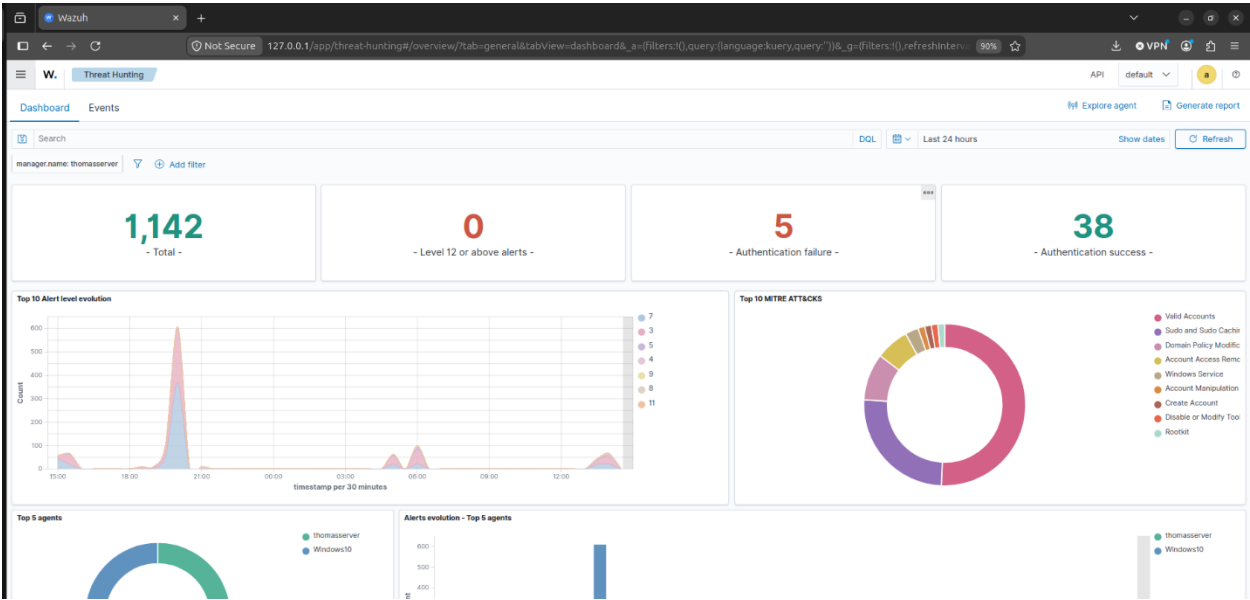
|                        |                     |                                   |
|------------------------|---------------------|-----------------------------------|
| Network Reconnaissance | Full port discovery | Generate detectable network event |
|------------------------|---------------------|-----------------------------------|

TABLE II - SIMULATED ATTACK SCENARIOS

# 6.RESULTS & OBSERVATIONS

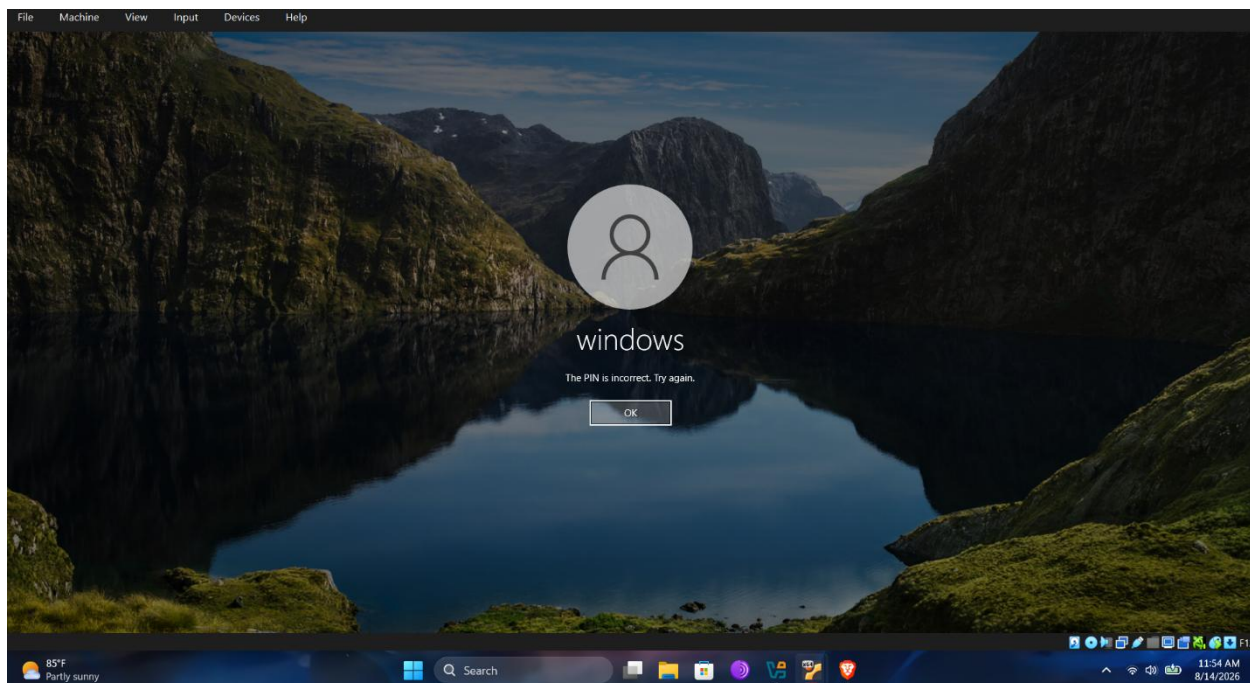
## A. Detection Results

The Windows agent enrolled successfully and remained active throughout testing. Security events and Security Configuration Assessment (SCA) results were visible in the dashboard. Fig. 5 presents the dashboard after the attack campaign, showing security events generated by the scanning and enumeration activity, while Fig. 6 shows authentication-failure events captured following invalid-credential attempts.



【FIGURE 05: After\_Dashboard.png】

Fig. 5. Wazuh dashboard after attack simulation, showing security events generated by scanning and enumeration activity.



【FIGURE 06: Invalid\_credentials.png】

Fig. 6. Authentication failure events captured on the Windows endpoint following invalid credential attempts.

## B. Observations

Network-based activity originating from Kali was partially visible at the endpoint, indicating that richer host telemetry (e.g., Sysmon) is required for stronger visibility. Core SOC L1 workflows were practiced, including dashboard navigation, event filtering by agent and source IP, review of rule groups and MITRE ATT&CK mappings [4], and basic alert triage. The principal dashboard modules used were Endpoints Summary, Threat Hunting, Security Events, Configuration Assessment (SCA), and MITRE ATT&CK.

## 7. Challenges Faced & Solutions

| Challenge                        | Solution                                                    |
|----------------------------------|-------------------------------------------------------------|
| Agent not appearing              | Verified network connectivity and agent configuration       |
| Limited visibility of nmap scans | Identified need for Sysmon for richer endpoint telemetry    |
| Network adapter inconsistencies  | Standardized all VMs to Host-only Adapter                   |
| Hardware resources limitations   | Adjusted VM resources and used ignore flags where necessary |

TABLE II – CHALLENGES AND SOLUTIONS

## 8. SKILLS GAINED AND FUTURE WORK

The project developed practical competence in SIEM deployment and administration, agent installation and troubleshooting, log analysis and alert triage, attack simulation, threat hunting fundamentals, virtual lab networking, and professional documentation. Planned extensions include installing Sysmon on the Windows endpoint; adding a Linux endpoint agent; creating custom Wazuh detection rules; simulating brute-force and privilege-escalation techniques; building custom dashboards; and deeper vulnerability-detection integration.

## 9. CONCLUSION

This paper presented a Wazuh-based SOC home laboratory providing hands-on experience in SIEM deployment, endpoint monitoring, attack simulation, and alert triage. The deployment validated the log collection, correlation, and visualization pipeline underlying SOC operations and demonstrated detection of simulated attacks including port scanning and authentication failures. The laboratory will be expanded with additional telemetry sources and advanced attack scenarios to further develop blue-team capabilities.

## 10. References

- [1] Wazuh, Wazuh Official Documentation. [Online]. Available: <https://documentation.wazuh.com>
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